

De Havilland Gipsy I

The first in a new series of engines developed for de Havilland, the Gipsy I was an upright, four-cylinder air-cooled engine with wet sump lubrication. It had a simple design, even for the standards of the 1920s, yet developed a respectable 85 hp with excellent reliability and fuel economy and was also easy to maintain.

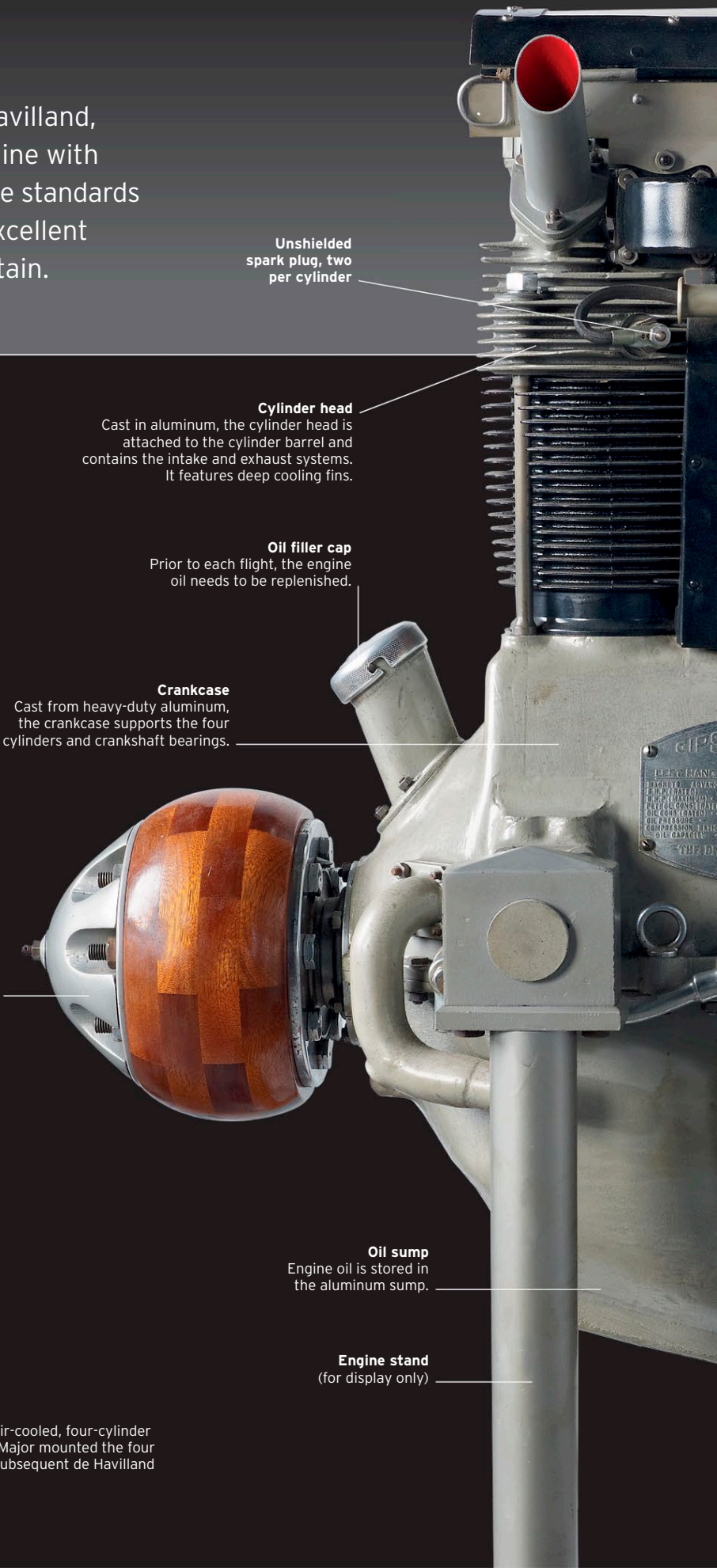
THE GIPSY SERIES

Major Frank Halford, a well-known engine design consultant, developed the Gipsy series of engines for de Havilland starting in the mid-1920s. The Gipsy I and Gipsy II were both used in de Havilland's Gipsy Moth, and perhaps the most famous application of the Gipsy I was that of powering Amy Johnson's de Havilland Moth on a world-record-breaking trip from England to Australia. Further development saw the Gipsy morph into the Gipsy Major, with inverted cylinders for better pilot vision and other improvements. Gipsy Majors were manufactured well into the years after World War II.

ENGINE SPECIFICATIONS	
Dates produced	1927 to mid-1930s
Configuration	Air-cooled 4-cylinder inline
Fuel	70 octane fuel
Power output	85 hp @ 1,900 rpm
Weight	285 lb (129 kg) dry
Displacement	318.1 cu in (5.21 liters)
Bore and stroke	4.5 in x 5 in (11.43 cm x 12.7 cm)
Compression ratio	5:1



▷ See Piston engines pp.302-303



Unshielded
spark plug, two
per cylinder

Cylinder head
Cast in aluminum, the cylinder head is
attached to the cylinder barrel and
contains the intake and exhaust systems.
It features deep cooling fins.

Oil filler cap
Prior to each flight, the engine
oil needs to be replenished.

Crankcase
Cast from heavy-duty aluminum,
the crankcase supports the four
cylinders and crankshaft bearings.

Propeller position
Although not present,
the propeller would
normally fit here.

Oil sump
Engine oil is stored in
the aluminum sump.

Engine stand
(for display only)

Inline layout
Although the Gipsy I had an air-cooled, four-cylinder
inline layout, the later Gipsy Major mounted the four
cylinders inverted, as did all subsequent de Havilland
aircraft piston engines.